DEPARTMENT OF CSE(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

DATABASE MANAGEMENT SYSTEMS LABORATORY

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EXERCISE –7: PROCEDURES **Date:10/09/2026**

1. Procedure to Find Palindrome of a Given Number

AIM:

To write and execute a PL/SQL procedure to check whether the given number is a palindrome or not.

ALGORITHM:

1. Start.
2. Read the given number NUM.
3. Store the original number in TEMP.
4. Initialize REV to 0.
5. Extract the last digit of N using MOD(N,10).
6. Add the extracted digit to REV after multiplying REV by 10.
7. Remove the last digit from NUM using TRUNC(NUM/10).
8. Repeat steps 5–7 until NUM becomes 0.
9. Compare REV with TEMP.
10. If both are equal, display Palindrome; otherwise, display Not a Palindrome.
11. Stop.

PL/SQL PROCEDURE:

```
SET SERVEROUTPUT ON;

CREATE OR REPLACE PROCEDURE PALINDROME(N IN NUMBER)
IS
   TEMP  NUMBER;
   NUM   NUMBER;
   REV   NUMBER := 0;
   DIGIT NUMBER;
BEGIN
   TEMP := N;
   NUM := N;

   WHILE NUM > 0 LOOP
      DIGIT := MOD(NUM, 10);
      REV := REV * 10 + DIGIT;
      NUM := TRUNC(NUM / 10);
   END LOOP;
```

```
    IF REV = TEMP THEN
       DBMS_OUTPUT.PUT_LINE(TEMP || ' is a Palindrome');
    ELSE
       DBMS_OUTPUT.PUT_LINE(TEMP || ' is Not a Palindrome');
    END IF;
END;
/
```

EXECUTION

EXEC PALINDROME(121);

OUTPUT

121 is a Palindrome

Another example:

EXEC PALINDROME(123);

Output:

123 is Not a Palindrome

Note:

EXEC means **execute (run) a stored PL/SQL procedure** in Oracle.

Here:

- EXEC → tells Oracle to execute/run the procedure
- PALINDROME → procedure name
- 121 → value passed to parameter N

You can also use the full PL/SQL syntax

- Instead of:
- `EXEC PALINDROME(121);`

you can write:

```
BEGIN
    PALINDROME(121);
END;
/
```

- Both execute the procedure.

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2. Procedure to Find GCD of Two Numbers

AIM

To write and execute a **PL/SQL procedure** to find the **Greatest Common Divisor (GCD)** of two given numbers using the **Euclidean algorithm**.

ALGORITHM

1. Start.
2. Read two numbers A and B.
3. Compare A and B.
4. If A < B, interchange A and B.
5. Divide the greater number by the smaller number.
6. Find the remainder using MOD(A,B).
7. If the remainder is 0, then B is the GCD.
8. Otherwise:

- Assign A = B.
- Assign B = remainder.

9. Repeat steps 5–8 until the remainder becomes 0.
10. Display B as the GCD.
11. Stop.

PL/SQL PROCEDURE:

```
SET SERVEROUTPUT ON;

CREATE OR REPLACE PROCEDURE GCD(
   A IN NUMBER,
   B IN NUMBER
)
IS
   X NUMBER := A;
   Y NUMBER := B;
   R NUMBER;
BEGIN
   -- Make X the greater number
   IF X < Y THEN
      R := X;
      X := Y;
      Y := R;
   END IF;

   -- Euclidean algorithm
   WHILE Y <> 0 LOOP
      R := MOD(X, Y);
      X := Y;
      Y := R;
   END LOOP;

   DBMS_OUTPUT.PUT_LINE('GCD = ' || X);
END;
/
```

EXECUTION:

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```
EXEC GCD(48, 18);
```

OUTPUT

GCD = 6

GCD Working Example

For 48 and 18:

48 ÷ 18 → Remainder = 12

18 ÷ 12 → Remainder = 6

12 ÷ 6 → Remainder = 0

Therefore, GCD = 6